



Reflection and E-Portfolio Report for Machine Learning Module

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Module: Machine Learning

GitHub Repository URL: <https://alisalem6062.github.io/E-Portfolio/machine-learning.html>

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First: Reflective Journey Of Machine Learning Module

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1.Introduction

Before this course, I knew about AI and ML from movies and news. I thought of robots, smart machines, and the future. Some stories made AI sound very good or very bad. In this course, I learned the real side of AI and ML. They are not magic. They are based on math and logic. People build them using data and rules. I saw that AI depends on human choices. It can be helpful or harmful. Each unit helped me understand more. Reflecting on my learning through Rolfe et al.'s (2001) model of “What?”, “So what?”, and “Now what?” has helped me digest the course’s intense technical content and consider how my personal values, learning strategies as well as aspirations evolved throughout the experience.

2. What? – The Learning Experience

From the first unit, I learned basics of machine learning. I found out that AI has many parts like machine learning, deep learning, and NLP. I learned the difference between narrow AI and general AI. Narrow AI does one task well. General AI can think like humans, but it doesn't exist yet. The course showed me simple ML algorithms. I learned about linear regression and classification. At first, the math was hard. I did not understand cost functions and gradient descent. But Unit 2 explained it well. The coding tasks helped me learn. I trained my first ML model. It predicted house prices. The results were not perfect, but I was happy. I saw how machines learn from data. It was an exciting moment.

As I went further in the course, I learned about unsupervised learning. It was harder to understand. Clustering with k-means showed me how machines find patterns without labels. I

used customer data and saw how the model grouped people by behavior. This was exciting. In Unit 7, I learned about anomaly detection. It is useful for spotting fraud and security issues.

In Unit 6, I studied deep learning and neural networks. It was tough at first. The layers and backpropagation were confusing. But the textbook by Bishop and Bishop (2024) helped. I saw how networks work like the brain. My “aha” moment came with the MNIST dataset. I trained a model to read handwritten digits. Seeing the accuracy improve felt amazing. It made AI feel real.

3. So What? – Why It Mattered

These lessons were more than just technical. The course changed how I see AI in society. It made me think about ethics and identity too. As a woman of African descent, Unit 12 stood out to me. It focused on bias and fairness in AI. I read a study by Buolamwini and Gebru (2018). It showed that facial recognition works worse for people with darker skin (Buolamwini et al., 2018). That upset me. I realized AI reflects the unfairness in our world. That moment changed how I think. I felt responsible not just as a tech person but as someone shaping AI. I started learning about fairness in machine learning. I looked into fixing data imbalance, checking for bias, and using models we can explain.

Using Schön’s (1983) idea of reflection-in-action, I started thinking about ethics while building models, not just after (Schön et al., 1983). I asked questions like: Who owns this data? Is it fair? What hidden ideas are in the model? In one task, we had to build and test a convolutional neural network. It worked well, but I didn’t know how it made decisions. This made me uneasy. So, I looked into Explainable AI (XAI). I used tools like SHAP and LIME to explain model results. This matched Lipton’s (2018) point that we need to understand models to trust them. I

learned that knowing why a model gives an answer is as important as the answer itself (Lipton et al., 2018).

Another important point was learning how AI affects public life. I saw how machine learning is used in courts, banks as well as jobs. These are areas where fairness matters most. Mehrabi et al. (2021) said that bias in AI is common and built into the system (Mehrabi et al., 2021). This made me think about using fair data. As a future worker, I want to speak up for equal treatment in AI.

4. Now What? – Moving Forward

This course changed how I see my future. I used to think machine learning was only technical. Now I see it's about people. The choices we make in algorithms and data affect real lives. AI is part of society. It reflects our values.

I now use what I learned in real projects. I'm working with healthcare data to predict disease risk. I think carefully at every step cleaning data, choosing models, checking results. I ask myself: Who will this affect? Is the data fair and diverse?

I am more committed to lifelong learning. Tools and libraries change often. But thinking clearly, ethically and with care will always matter. I now read academic journals and AI ethics blogs. I read work by Dastin (2018) and others. They talk about social effects of AI (Dastin, 2018).

As a learner, I ask myself three questions from Rolfe et al. (2001): What happened? Why does it matter? What will I do differently? This helps me overcome imposter syndrome, handle coding stress as well as reflect on both technical and social issues.

5. Conclusion

This course on machine learning and artificial intelligence was more than just learning skills. It became a personal journey for me. I didn't just learn how to build smart systems. I learned to think deeply about how these systems affect people and society. I used the reflection model by Rolfe et al. (2001) to guide my thinking. It helped me look at what happened, what I learned, and what I will do next. Schön's (1983) ideas reminded me that learning is not a one-time event. It keeps going, and we must reflect as we grow. During this course, I became more confident. I can now understand and work with complex systems. I feel more ready to face things that are not always clear or certain. I now see that learning AI and ML is not only about writing code. It's about using these tools in a responsible way. AI is a powerful and life-changing technology. I feel ready to be part of it, with both skill and care.

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Second: E-Portfolio Report

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1. Introduction

This report explains my experience of making an individual e-Portfolio for the Digital Ethics and Technological Literacy module at the University of Essex. I built the portfolio on GitHub. It shows my weekly work with digital tools, especially Generative Artificial Intelligence (GenAI). I explored the ethical issues that come with using these tools in school and work. Each week, I did tasks like testing AI tools, learning about academic honesty, and improving my digital skills. I used markdown to write down what I learned. I reflected on my thoughts and linked them to ideas from academic studies.

2. Description of the e-Portfolio Work

My e-Portfolio is on GitHub. It is organised into folders for each week (Week 1 to Week 12). Each folder has markdown files. These files show what I learned, the tasks I did, the AI tools I used, and my thoughts about the week. The main page has a README.md file. It explains the purpose of the e-Portfolio. It helps users understand how to move through the folders.

Each week focused on a different aspect of GenAI ethics, including:

- Understanding the foundations of AI and machine learning (Week 2)
- Prompt design and output critique (Week 3)
- Evaluation of bias in AI-generated imagery (Week 5)
- Ethical dilemmas in using AI for academic writing (Week 6)
- Open discussion about the future of AI and digital citizenship (Week 9)

I took screenshots and tested different prompts to show how I used AI tools. I shared the pictures and code results. To keep track of my changes, I used Git. This helped me stay organised and show how I improved over time.

The portfolio provided a space to explore tools like:

- ChatGPT for language-based tasks
- Canva's AI text-to-image tool for visual experiments
- Google Bard for comparison with other LLMs
- GitHub Copilot for examining AI-assisted code generation

This multi-tool approach helped me understand both the capabilities and constraints of current GenAI systems.

3. Key Learning Outcomes

3.1 Enhancing Digital Competence

Before this module, I didn't know how to use GitHub. I had never made a digital portfolio with version control. But with step-by-step lessons and practice, I learned the basics. I learned how to start a repository, write and format markdown files, use Git to push changes, and keep my folders organised.

My confidence with GenAI tools got better. In Week 3, I used Google Bard to make summaries of the course content. I compared them with what ChatGPT gave me. This helped me see how different AI tools give different answers. I got better at writing prompts that give more useful results. These skills will help me in school and in my future job. AI tools are becoming a big part of how we work and study.

3.2 Developing Ethical Awareness

One big theme in this module was learning about the ethics of GenAI. In Week 6, we used ChatGPT to write a reflective paragraph and compared it to our own writing. The AI text was

clear, but it did not show real feelings or deep thoughts. This showed me that human reflection is special and something AI cannot truly copy.

Floridi and Chiriatti (2020) said that GenAI does not really understand what it writes. They explained that AI can make sentences that look correct but have no real meaning. I saw this when I asked ChatGPT to help with a debate about privacy rights. The arguments made sense, but they felt empty. They did not include real life experiences or human values.

This made me ask more questions. Can GenAI ever be fair or neutral? Who is to blame if AI causes harm? How do power and privilege affect the data that trains AI? These questions helped me learn more about AI ethics. They showed me how important it is for humans to always guide and watch over AI tools.

3.3 Reconsidering Academic Integrity

In Week 7, I thought a lot about using GenAI tools like ChatGPT to help with my studies. Sometimes, I felt tempted to use it for things like paraphrasing or generating ideas. It seemed like a quick and easy way to get work done. But I realised that if I relied on it too much, or didn't say I used it, it could hurt my learning. It might go against academic rules.

The University of Essex's Academic Integrity Policy (2023) helped me understand the right way to use AI tools. After reading it, I decided to be honest and open whenever I used AI. I started clearly writing down when I used a tool like ChatGPT. I explained what the AI helped me with and how I changed the answer in my own words.

For example, in Week 8, I used ChatGPT to summarise an article about ethics. I didn't just copy the summary. Instead, I read it, tried to understand it, and then rewrote it using my own

ideas. In my work, I gave credit to both the original article and the AI tool. This way, I followed academic rules and showed my own understanding.

The e-Portfolio helped me think more deeply about how to use AI. I learned that AI should not replace my thinking. Instead, it should support my learning. It helped me see the value of honesty, editing, and using my own voice. This new mindset made me feel more confident and responsible in my work.

4. Ethical Reflection and Use of Generative AI

Reflecting on the ethical dimensions of GenAI use, I identified several key tensions:

4.1 Bias and Representation

In Week 5, I used Canva's text-to-image AI to make pictures of "a successful scientist." Most of the images showed men who looked Western. This made me think about how AI can repeat old ideas and stereotypes.

I tried writing different prompts to get more diverse images. I learned that we need to ask AI the right way if we want it to show more people from different backgrounds. Using AI in a fair way means we must think carefully about what we ask it to do.

This made me wonder: Who gets shown in AI images? Who gets left out? If the data used to train AI is not diverse, some voices may be erased. Now I think that using AI in a good way needs more than just honesty. It needs care and effort to include everyone.

4.2 Accountability and Attribution

Another important issue was authorship. If an AI tool helps with an assignment, who is the real author? Some schools are starting to make rules about this. But for now, it is mostly up to each

person to be honest. In my portfolio, I chose to be open about using AI. I wrote down clearly where I used AI and what parts I changed or edited. I added footnotes to explain this. I think being honest like this builds trust. It shows that I care about doing the right thing in school and work. AI tools are not good or bad by themselves. What matters is how we use them.

5. Future Applications and Development

This module and e-Portfolio experience have changed how I view AI and digital ethics, not only in academic settings but in broader societal contexts.

5.1 Academic Practice

I will continue using GenAI tools for brainstorming, editing, and feedback but always with a critical lens. I plan to:

- Disclose AI use in footnotes or appendices.
- Avoid using AI for content creation in graded assignments.
- Use AI-generated outputs as learning tools rather than final submissions.

I intend to apply what I have learned when assisting others, such as peers who may struggle with academic writing or digital skills. I believe in ethical peer support that empowers rather than shortcuts the learning process.

5.2 Career and Professional Life

As I prepare for a career in communications, understanding how to responsibly use GenAI is crucial. I see practical applications in:

- Content drafting and ideation
- Automated summarisation of reports

- Visual design assistance

However, I am now more aware of the need to assess bias, verify sources, and maintain a human editorial role. In fast-paced industries, the temptation to delegate too much to AI is strong—but so are the ethical consequences of doing so.

I plan to attend webinars or short courses on AI literacy, AI in business ethics, and the legal aspects of automation. These will help deepen my knowledge and keep me updated in a rapidly evolving field.

5.3 Commitment to Digital Citizenship

I want to be a good and honest digital citizen. I will speak up for fair and open use of AI. I believe AI should treat everyone equally. I want to support clear and honest rules for using AI. As a student, worker, or community member, I will follow these values. I want to help share free learning tools. This will help more people understand digital skills and AI.

6. Conclusion

Creating and updating my e-Portfolio was both a technical and personal experience. I learned how to use digital tools better. At the same time, I started thinking more deeply about how to use GenAI in a fair and honest way.

I faced some important questions. These included how to deal with bias in AI, who should get credit for work, and how to stay honest in my studies. I thought about how open I should be when I use AI. These challenges helped me understand that using AI needs care and responsibility.

This report and my GitHub page show the skills I have learned. They reflect my growth in using AI in a smart, careful, and fair way. I now feel more ready to use AI in school and at work. Most of all, I want to use these tools in a way that is honest, fair, and includes everyone.

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